Study Notes: Data Analysis, ML Project Planning & Tensors

1. Short Notes

a. What is Data Analysis?

- **Definition:** The systematic process of inspecting, cleaning, transforming, and modeling raw data to discover useful patterns, draw conclusions, and support business decision-making.
- **Core Stages of the Analysis Lifecycle:**
    1. **Data Collection:** Gathering raw inputs from databases, APIs, web scraping, or logs.
    2. **Data Cleaning (Wrangling):** Removing duplicates, treating missing values, correcting data types, and filtering anomalies.
    3. **Exploratory Data Analysis (EDA):** Computing summary statistics and plotting distributions/correlations to reveal underlying structure.
    4. **Data Modeling & Interpretation:** Applying statistical or machine learning models to identify relationships.
    5. **Data Visualization & Reporting:** Presenting actionable takeaways using charts, dashboards, and reports.
- **Primary Types:**
    - **Descriptive:** *What happened?* (e.g., historical sales summaries)
    - **Diagnostic:** *Why did it happen?* (e.g., drill-down churn root cause)
    - **Predictive:** *What is likely to happen?* (e.g., forecasting next quarter's demand)
    - **Prescriptive:** *What action should be taken?* (e.g., route optimization)

b. How to Plan a Data Science Project

- **Step 1: Business Understanding & Objective Definition**
    - Define the business problem clearly (e.g., *"Reduce user churn by 5%"* rather than *"Build an AI model"*).
    - Determine project KPIs and criteria for success.
- **Step 2: Feasibility & Data Discovery**
    - Identify data availability, quality, privacy constraints, and access permissions.
    - Estimate computing resources and engineering overhead.
- **Step 3: Roadmap & Milestone Formulation**
    - Break the project into phases: Data Ingestion → EDA → Baseline Modeling → Model Iteration → Deployment.
    - Establish timelines and deliverable checkpoints using agile frameworks (e.g., CRISP-DM).
- **Step 4: Tooling & Environment Setup**

    - Select languages, libraries, storage (data lake/warehouse), and version control systems (Git, DVC).

- **Step 5: Risk Management & Evaluation Strategy**
    - Establish an offline evaluation benchmark (e.g., hold-out test set, cross-validation).
    - Plan for edge cases, label shift, and model degradation over time.

c. How to Frame a Machine Learning Problem

- **1. Translate Business Objective to ML Task:**
    - Map requirements to a concrete learning paradigm:
        - **Supervised:** Regression (predicting a continuous value) vs. Classification (predicting discrete labels).
        - **Unsupervised:** Clustering, Dimensionality Reduction, Anomaly Detection.
        - **Reinforcement:** Agent learning policy via rewards/penalties.
- **2. Define Target & Feature Space:**
    - **Input Features (X):** Measurable signals available at inference time without data leakage.
    - **Target Label (y):** Explicitly ground truth definition and acquisition method.
- **3. Select Metric Aligned to Business Value:**
    - Choose optimization and evaluation metrics:
        - *Classification:* Accuracy, Precision, Recall, F1-Score, ROC-AUC (e.g., prioritize Recall for disease detection).
        - *Regression:* MAE, MSE, RMSE, MAPE.
- **4. Define the Serving Architecture:**
    - Batch inference (offline scheduled jobs) vs. Real-time inference (low-latency REST/gRPC API).
- **5. Establish a Strong Baseline:**
    - Build a simple rule-based heuristic or linear model before implementing complex architectures.

2. In-Depth Explanation: Tensors (with NumPy Examples)

What is a Tensor?

- In computing and machine learning, a **tensor** is a generalized mathematical container for numerical data.
- It extends the concept of scalars, vectors, and matrices to arbitrary dimensions (referred to as **axes** or **ranks**):
    - **Rank-0 Tensor:** Scalar (single number)

- **Rank-1 Tensor:** Vector (1D array)
- **Rank-2 Tensor:** Matrix (2D grid of numbers)
- **Rank-3+ Tensor:** Multi-dimensional array (volumes, sequences, image batches, video frames)

Key Attributes of a Tensor:

1. **Rank / Number of Axes (ndim):** How many indices are needed to reference a specific value.
2. **Shape (shape):** A tuple of integers representing the size along each axis.
3. **Data Type (dtype):** The data format stored within the tensor (e.g., float32, int64).